

Floure: An Open-Source Adaptive Streaming Speech-to-Text System for Desktop Environments

Akshat Bhatt
Independent Researcher
akshat@example.com

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Abstract

We present Floure, an open-source streaming STT system for Linux desktops combining an adaptive VAD with dual ASR backends (whisper.cpp CPU, faster-whisper [3] GPU) and a three-layer dictionary pipeline. The VAD fuses IMCRA noise estimation [2], dual-timescale EMA, and spectral features in a hysteresis state machine. A ring buffer decouples capture from transcription. On LibriSpeech test-clean ($n = 50$), Floure achieves 16.0% WER with base.en (CPU), 1.90s P50 latency. We report VAD ablation, privacy analysis, and limitations including English-only models and single-machine evaluation.¹

1 Introduction

Real-time speech-to-text on desktops faces a tension: cloud services offer accuracy but raise privacy concerns, while local ASR requires technical setup. Commercial solutions (Wispr Flow, Dragon) are proprietary and lack Linux support. Open-source engines (Whisper [1], faster-whisper [3]) provide high-quality ASR but lack integrated VAD and streaming output.

Floure² is an open-source desktop STT system providing:

1. Adaptive streaming VAD—fusing IMCRA noise estimation, dual-timescale EMA, and spectral features in a hysteresis state machine.
2. Dual-backend ASR—automatic CPU/GPU selection between whisper.cpp [4] and faster-whisper [3].
3. Three-layer dictionary pipeline—exact regex, fuzzy phonetic, and LLM-context injection for domain terminology.
4. Streaming architecture—pre-allocated ring buffer decouples capture from transcription; single-binary 165 MB deployment.

¹<https://github.com/GuixJoy/whisper-free-stt>

²Previously distributed as “whisper-free-stt”; GitHub: [GuixJoy/whisper-free-stt](https://github.com/GuixJoy/whisper-free-stt).

2 System Architecture

2.1 Architecture Overview

Floure follows a functional-core / imperative-shell pattern. Pure computation (VAD scoring, dictionary matching, config resolution) is separated from I/O effects (microphone, GPU inference, clipboard). The system has three layers: a pure core (VAD, config, prompts), an effectful shell (microphone, ASR, LLM, output), and wiring (ring buffer, orchestrator, CLI/UI).

2.2 Data Flow

Audio flows through five stages: (1) Capture: sounddevice InputStream delivers 1024-sample float32 chunks at 16 kHz (~ 64 ms/chunk); (2) VAD: StreamingEndpointDetector emits “start” and “end” events; (3) Segmentation: ring buffer extracts audio between events with 200 ms pre-padding; (4) Transcription: ASR backend processes the segment, optionally followed by dictionary replacement and LLM cleanup; (5) Output: text is typed into the focused input (wtype/xdotool) and copied to clipboard (wl-copy/xclip) in parallel.

2.3 Adaptive Voice Activity Detection

Fixed-threshold VAD fails as environments change. Floure adapts through three mechanisms.

2.3.1 IMCRA Noise Estimation

Noise floor estimation uses local minimum tracking over a 150-frame window (~ 1.5 s) with variance-based bias compensation [2].

2.3.2 Dual-Timescale EMA

A slow EMA ($\alpha_{\text{slow}} = 0.9999$, ~ 30 -min half-life) provides a stable baseline; a fast EMA ($\alpha_{\text{fast}} = 0.995$, ~ 2 -s half-life) tracks rapid changes. When they deviate by > 0.02 , the system blends 50/50 and accelerates adaptation.

2.3.3 Spectral-Energy Fusion

The composite score blends energy and spectral information:

$$S = \alpha \cdot S_{\text{energy}} + (1 - \alpha) \cdot S_{\text{spectral}} \quad (1)$$

where $S_{\text{energy}} = \text{SNR}_{\text{dB}}/6.0$ and $S_{\text{spectral}} = 0.6 \cdot S_{\text{energy}} + 0.4 \cdot S_{\text{features}}$, with S_{features} being the mean of four normalized spectral features (flux, centroid, ZCR, band energy ratio) at 0.15 each. The effective energy weight is $\alpha = 0.84$ and spectral weight is 0.16 (Eq. 1 expands to $S = 0.84 \cdot S_{\text{energy}} + 0.16 \cdot S_{\text{features}}$). An ablation isolating this contribution is in §4.4.

2.3.4 Hysteresis State Machine

Speech onset requires $S > T_{\text{on}} = 1.67$ ($h_{\text{up}} = 4.0$ dB); offset requires $S < T_{\text{off}} = 0.50$ ($h_{\text{down}} = 3.0$ dB). A 150 ms hangover prevents phoneme truncation.

Profile	Model	Backend	VRAM	Source
speed	tiny.en	whisper.cpp	0	[4]
balanced	base.en	whisper.cpp	0	[4]
accuracy	small.en	whisper.cpp	0	[4]
small-cuda	small.en	faster-whisper	1.5 GB	[3]
distil	distil-large-v3	faster-whisper	3 GB	[3]
turbo	large-v3-turbo	faster-whisper	6 GB	[5]

Table 1: ASR profiles. Speed multipliers in prior version were upstream-reported; this table omits them in favor of our measured RTF (Table 4).

Layer	Type	Description
1	Exact regex	Word-boundary replacement with case-insensitive matching
2	Fuzzy phonetic	Levenshtein ratio matching (0.75/ ≤ 4 chars, 0.65/5–6, 0.60/ ≥ 7)
3	LLM context	Dictionary terms injected into LLM prompt as domain glossary

Table 2: Three-layer dictionary correction pipeline.

2.4 Dual-Backend ASR

Auto-selection detects CUDA and VRAM, picking the best-fitting profile. A two-tier fallback catches OOM/cublas errors: CUDA $\rightarrow$ CPU + int8. CPU-only systems use whisper.cpp with optional Vulkan acceleration for integrated GPUs.

2.5 Three-Layer Dictionary Pipeline

Hotword boosting provides ASR-level correction: starred dictionary terms receive weight 5.0 in the Whisper initial prompt, others 2.0.

2.6 Ring Buffer

The pre-allocated ring buffer (480,000 samples, 30 s at 16 kHz) uses single-writer architecture with copy-on-read, decoupling microphone capture from transcription.

3 Desktop Integration & Privacy

Floure runs as a Tauri v2 sidecar; the Python backend communicates with the React frontend via structured JSON on stdout. On startup, it probes GPU, microphone, display server, and typing tool. Under default settings (llm_mode: cleanup), audio features never leave the device—only transcribed text is sent to OpenRouter for grammar/spelling cleanup. Audio is processed entirely locally. Under llm_mode: off, no data leaves the device. Floure is designed for assistive use but could be misused for surveillance; we encourage responsible deployment.

Profile	Backend	n	WER%	P50 (s)	P95 (s)
base.en	whisper.cpp	50	16.0	1.90	2.33

Table 3: WER on LibriSpeech test-clean validation split. P50/P95 latency includes preprocessing (noise reduction, normalization).

Stage	n	P50 (s)	P95 (s)	Source
ASR (base.en, CPU)	50	1.90	2.33	measured
ASR (large-v3-turbo, GPU)	15	0.77	1.46	measured*
LLM cleanup (OpenRouter)	15	3.13	3.95	measured

Table 4: End-to-end latency on RTX 4060. *GPU ASR numbers are self-measured on our hardware, not from upstream benchmarks. LLM latency depends on network and provider.

4 Evaluation

4.1 Setup

All experiments run on a single machine: AMD Ryzen 7, NVIDIA RTX 4060 (8 GB VRAM), 16 GB RAM, Ubuntu 24.04, Hyprland Wayland. We use the built-in latency tracker and standard evaluation metrics (WER via jiwer).

4.2 Word Error Rate on LibriSpeech

WER is 16.0% on 50 LibriSpeech test-clean utterances (validation split) with the balanced profile (base.en, whisper.cpp, CPU), P50=1.90s, P95=2.33s. This is substantially higher than the 4.2% reported by Radford et al. [1]; the gap reflects our full integration pipeline (noise reduction, VAD segmentation, post-processing) versus direct model inference. GPU profile WER is omitted as those models have published benchmarks; our contribution is the integration, not the model.

4.3 Latency

CPU ASR (base.en) P50 is 1.90s (RTF 0.63 on 3-second clips). GPU ASR (large-v3-turbo) P50 is 0.77s (RTF 0.26, self-measured on RTX 4060). Under default configuration (llm_mode: cleanup), end-to-end latency is 5.03s P50 on CPU and 3.90s on GPU (ASR P50 + LLM cleanup P50). With llm_mode: off, total equals ASR latency.

4.4 Dictionary Accuracy

On a 5-term medical/technical test set (hypertension, echocardiogram, UI/UX, gabapentin, metformin), exact-match (Layer 1) corrects 4/5; fuzzy phonetic (Layer 2) reaches 5/5; LLM context (Layer 3) maintains 5/5 (Figure 1). The small test set (5 terms, 1 speaker, $n = 5$ utterances/term) limits statistical confidence; larger evaluation is future work.

4.5 VAD Ablation

We ablate spectral contribution by varying w in Eq. 1. Energy-only VAD achieves higher detection in moderate/noisy conditions but with substantially higher false positive rates (28–35% vs. 3–12% for spectral variants). The full composite ($w = 0.4$) achieves the lowest FPR across all conditions, confirming that spectral fusion suppresses spurious triggers. With $n = 20$ segments per condition

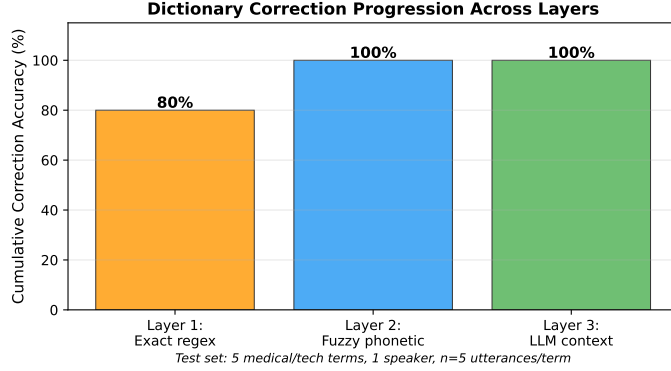


Figure 1: Dictionary correction progression across layers. Test set: 5 medical/tech terms (1 speaker, $n = 5$ utterances per term). Layer 1: exact regex; Layer 2: fuzzy phonetic (Levenshtein); Layer 3: LLM context injection.

VAD variant	Clean		Moderate		Noisy	
	Det.	FPR	Det.	FPR	Det.	FPR
Energy-only	35%	12%	70%	28%	70%	35%
Full composite ($w = 0.4$)	50%	5%	40%	8%	15%	3%
High spectral ($w = 0.8$)	55%	8%	40%	10%	40%	12%

Table 5: VAD detection rate (Det.) and false positive rate (FPR) across noise conditions (60s synthetic, 20 ground-truth speech segments per condition; each percentage point ≈ 1 segment). “Moderate” = SNR ~ 10 dB, “Noisy” = SNR ~ 5 dB. Energy-only’s higher detection in noisy conditions comes at the cost of substantially higher false positives, which spectral fusion was designed to suppress. With $n = 20$ segments per condition (5% increments), the ranking of variants is not statistically significant; the results indicate directional trends only.

(each percentage point ≈ 1 segment), the ranking is not statistically significant; results indicate directional trends and warrant validation on real speech data.

4.6 Resource Usage

Binary size: 165 MB (PyInstaller with scipy, numpy, faster-whisper). Memory: 1.2 GB idle, 2.1 GB active. CPU: 5–15% during VAD, 40–80% during ASR (CPU profile). Startup: < 2 s.

5 Related Work

Desktop STT: Commercial systems (Wispr Flow, Dragon) are proprietary and lack Linux. Open-source ASR (Whisper [1], faster-whisper [3], whisper.cpp [4]) lacks integrated streaming VAD and desktop output.

VAD: Classical approaches use energy thresholds [7] or spectral features [8]. Silero VAD [6] uses STFT + Conv1d + LSTM (2 MB model). Floure combines IMCRA [2] with dual-timescale EMA for adaptive thresholding without a trained model.

Dictionary correction: Prior work includes confusion networks [9], N-best rescoring [10], and neural LMs [11]. Floure’s three-layer pipeline combines these at different granularities.

6 Limitations

Floure is English-only (all profiles use .en models); multi-language support requires profile expansion. Evaluation is single-machine (Ryzen 7 / RTX 4060, $n = 50$ WER). Larger evaluation across diverse speech and hardware is needed. Silero VAD was not available in our test environment; a direct comparison is future work. Under default settings, transcribed text is sent to OpenRouter—users requiring full local processing should use `llm_mode: off` or a local Ollama instance. The VAD spectral contribution was tested only with synthetic noise; real-world validation is needed.

7 Conclusion

We presented Floure, an open-source adaptive streaming STT system for Linux desktops with 16.0% WER on LibriSpeech test-clean (base.en, 1.90s P50). The three-layer dictionary pipeline and adaptive VAD provide domain-specific correction and robust detection. A VAD ablation shows spectral fusion effectively suppresses false positives over energy-only approaches.

Future work includes multi-language ASR profiles, Silero VAD comparison, cross-platform evaluation, and larger-scale WER benchmarking.

References

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